

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0288

Ironwood Llc 345 kV
October 15, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0288. The Interconnection Customer proposes a 169.0 MW wind facility interconnecting at the Ironwood Llc 345 kV substation in the Pennsylvania area. The request was received on November 16, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 154 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,074,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Fayette to Mercury 345kV, the Grizzly to Del Cero 138kV loads to 113.2% of its 164 MVA rating. Project DFAX: 6.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grizzly Substation to Del Cero 138kV Line	Rebuild line with higher-capacity conductor	\$3,074,000	100% (\$3,074,000)
	Total allocated to project		\$3,074,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.